

1(a)(i)	<u>0.13</u> ;	1
1(a)(ii)	(reaction time would be) unchanged ; or (reaction time would be) longer / AW ;	1
1(b)(i)	distance ruler, drops / falls ;	1
1(b)(ii)	<i>any 7 of:</i> 1. test in silence / quiet / AW and noise / AW ; 2. method to ensure silence ; 3. ensure consistent / similar / same / known, level / volume / loudness / decibels, of noise / AW ; 4. suitable number of people tested ; 5. <i>idea of</i> no warning given when ruler to be dropped ; 6. <i>detail of ruler use:</i> same, ruler / starting distance / type or weight of ruler / held vertically ; 7. same or similar age / same sex ; 8. use same / dominant, hand ; 9. no, (named) drugs / medications / stimulants / depressants ; 10. <i>idea of</i> avoiding people with conditions affecting reaction time e.g. poor hearing / poor sight / neurological disorders ; 11. same, time of day / stated time ; 12. <i>ref. to</i> a minimum of three repeats and calculate a mean ; 13. low / medium, risk ;	7
1(c)	(independent variable / sound present or absent, is) discontinuous / discrete / qualitative / categoric, variable ;	1
1(d)(i)	<i>absence of noise</i> (mean) reaction time is longer or <i>presence of noise</i> (mean) reaction time is shorter ;	1
1(d)(ii)	comparing (two) <u>means</u> / continuous data / normal distribution ;	1
1(d)(iii)	there is no (significant) difference between the (mean) reaction times for no (background) noise and (background) noise ;	1

Question	Answer	Marks
1(d)(iv)	not significant / is due to chance (at $p > 0.05$) and because, it / calculated value / 2.05, is less than, the critical value of t / table value of t / 2.12 ; ora	1
1(e)(i)	<i>idea of</i> the (number of), repeats (of the task) / AW ;	1
1(e)(ii)	<i>idea that</i> as repeats increase, (number of) errors decrease / accuracy increases ;	1
1(e)(iii)	(student Y , replicate 5) – circle around the number 44 ;	1
1(e)(iv)	<i>idea of</i> (student) Y and as, had fewer errors / were more accurate (than rest of students) ; or as (results) start off lower / as (results) are lower in trial one / as even from the very beginning (results) are lower ;	1
1(f)	<i>any 2 of:</i> <i>idea that</i> the more, replicates / repeats, carried out, fewer errors are made ; <i>idea of</i> after, 6–7 / several trials, number of errors starts to plateau / number of errors does not continue to decrease ; <i>idea that</i> interruptions / AW, increases number of errors ;	2